



Chapter 17

Binocular Vision

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Outline

- The flowchart to build a binocular system
- Rectified binocular system
- Calibration of a physical binocular system
- Establishment of the rectified binocular system
- Obtaining the rectified binocular pair from a physical binocular pair
- Stereo matching and disparity map computation for a rectified binocular pair
- 3D reconstruction based on the disparity map in the rectified binocular system
- Practice



The flowchart to build a binocular system

online

offline

Based on the disparity map, compute the depth map and the final point cloud

With the rectified binocular pair, perform stereo matching and disparity map computation

For an input binocular pair, obtain their projections in the rectified binocular system (rectified binocular pair)

Establish the rectified binocular system

Calibrate the binocular system

For the pixel $\mathbf{u}_l$ on the left rectified image, if its disparity is d , then the z -value of its associated spatial point is $f_{rect} b / d$ and finally the 3D position of this spatial point can be obtained (w.r.t the camera system of the left rectified camera)

First, establish the pixel matching relationship between the rectified binocular images. For a matching pixel pair $\mathbf{u}_l = (x_l, y)$ and $\mathbf{u}_r = (x_r, y)$ on the left and right rectified images, the disparity for $\mathbf{u}_l$ is $x_l - x_r$

At runtime, for an input binocular pair, obtain their projections in the rectified binocular system, i.e. the rectified binocular pair

The rectified binocular system is a “virtual” system, derived from the physical binocular system; in such a system, the left-right two images are **line-aligned** row by row

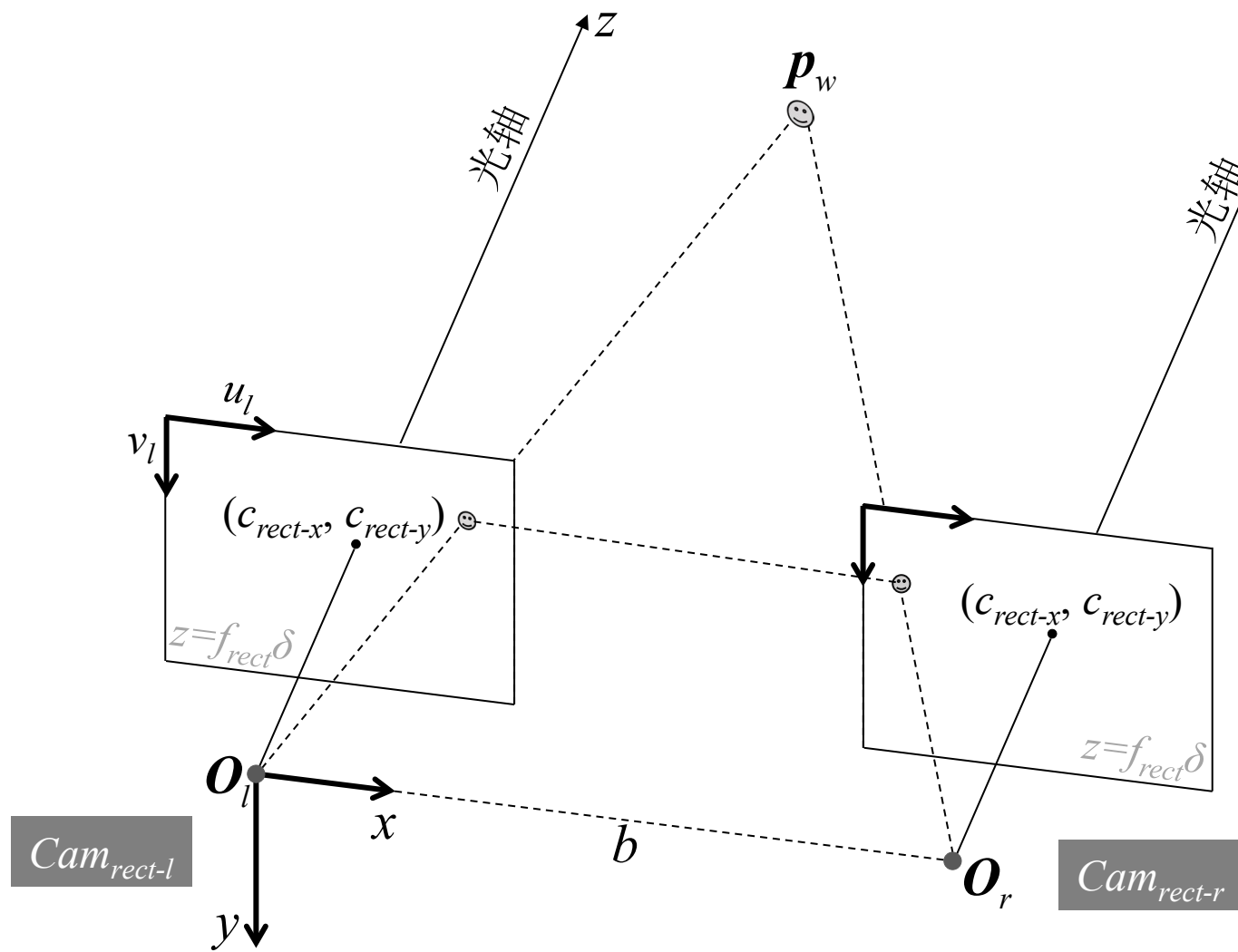
Get the intrinsics of the two cameras and their extrinsics, $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ and $\mathbf{t} \in \mathbb{R}^3$



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Rectified binocular system



- f_{rect} is the focal length (unit: *pixel*) of the intrinsics matrix
- δ is the physical length of each pixel (unit: *mm/pixel*)
- (c_{rect-x}, c_{rect-y}) is the position of the principal point on the imaging plane (unit: *pixel*)
- b (unit: *mm*) is the distance between the two camera centers

For the rectified binocular system, the distortion coefficient vectors for the two cameras are **0s**



Rectified binocular system

- Rectified binocular system is a “**virtual**” system; it can simplify the depth estimation
- By moving $\mathbf{O}_l\text{-}xyz$ $b(mm)$ along the x-axis, we get the $\mathbf{O}_r\text{-}xyz$ coordinate system
- Cam_{rect-l} and Cam_{rect-r} have the same intrinsics matrix,

$$\mathbf{K}_{rect} = \begin{bmatrix} f_{rect} & 0 & c_{rect-x} \\ 0 & f_{rect} & c_{rect-y} \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$



The two images $\mathbf{I}_{rect-l}$ and $\mathbf{I}_{rect-r}$ of a rectified binocular system are row-aligned. That means, if $\mathbf{u}_l(u_l, v_l)$ and $\mathbf{u}_r(u_r, v_r)$ are the images of the same spatial point on $\mathbf{I}_{rect-l}$ and $\mathbf{I}_{rect-r}$, we should have $v_l=v_r$



An assignment



Rectified binocular system

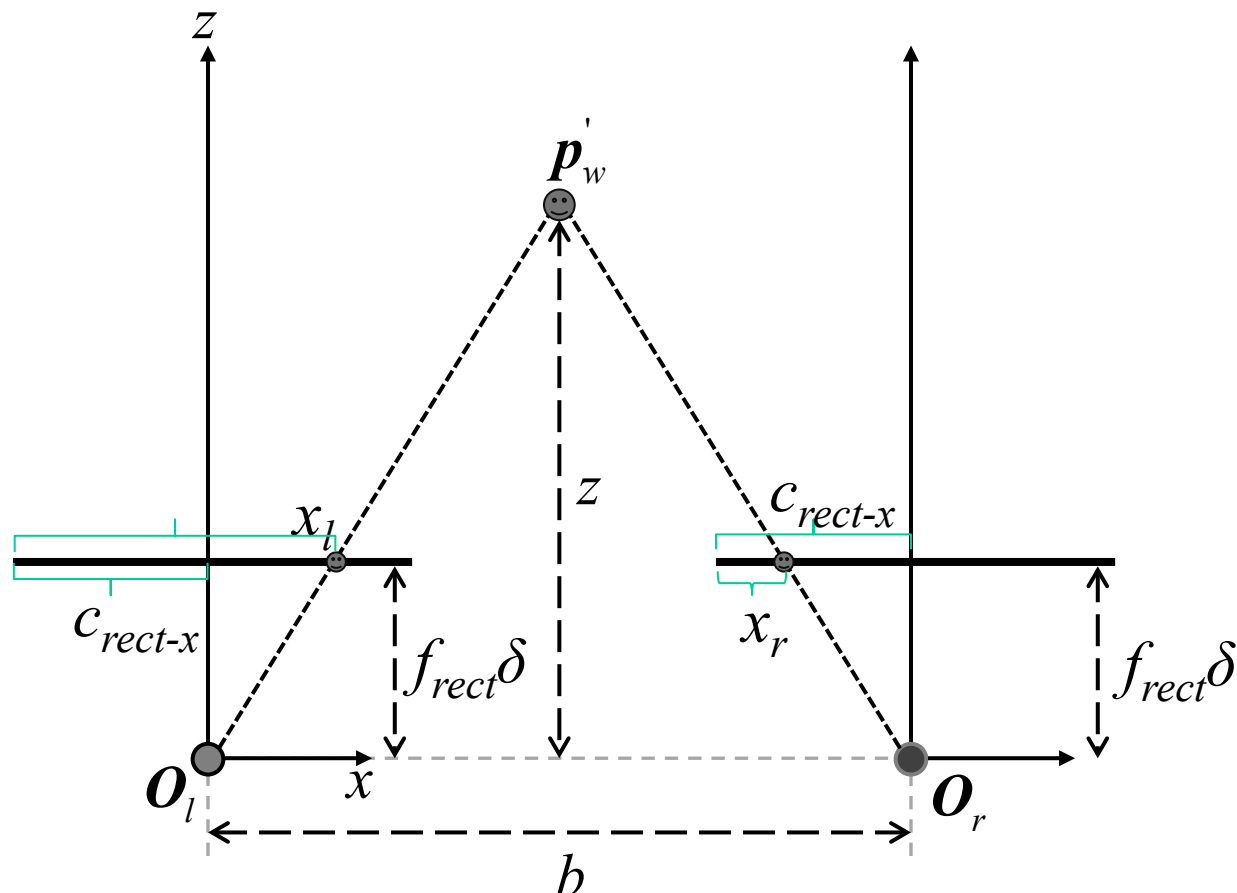
- Depth calculation under the rectified binocular system
 - With the rectified binocular system, for a spatial point p_w , if we know **its projections on the two rectified images**, it is easy to obtain its **depth w.r.t. to the left rectified camera**, i.e., p_w 's z-value with respect to the coordinate system $O_l\text{-}xyz$
 - Suppose that p_w 's image on I_{rect-l} is $u_l(x_l, y)$, and its image on I_{rect-r} is $u_r(x_r, y)$
 - Projecting all the elements on the plane $O_l\text{-}xz$. On $O_l\text{-}xz$, the projection of p_w is p'_w . Obviously, p'_w 's z-value w.r.t. to the coordinate system $O_l\text{-}xz$ is p_w 's z-value with respect to the coordinate system $O_l\text{-}xyz$





Rectified binocular system

- Depth calculation under the rectified binocular system



The **physical distance** between two image pixels $u_l(x_l, y)$ and $u_r(x_r, y)$ is,

$$b - [(x_l - c_{rect-x}) + (c_{rect-x} - x_r)] \delta = b - (x_l - x_r) \delta$$

Then we have,

$$\frac{z - f_{rect} \delta}{b - (x_l - x_r) \delta} = \frac{z}{b} \Rightarrow z = \frac{f_{rect} b}{x_l - x_r} \quad (2)$$

Disparity of u_l

To get the depth map associated to I_{rect-l} , we need to get the disparity map of I_{rect-l}

With the rectified binocular system, to compute the depth of a 3D point p_w with respect to O_l -xyz, only the disparity of its image u_l needs to be known



Rectified binocular system

As we have said, the rectified binocular system is a “virtual” system

Given a real physical binocular system, how to get its associated rectified version?



We need to perform calibration to the physical binocular system; with its extrinsics and the intrinsics known, its rectified version can be derived



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Calibration of a physical binocular system



To calibrate a physical binocular system

Calibrate the intrinsics of the two cameras

We have learned in Chapter 10

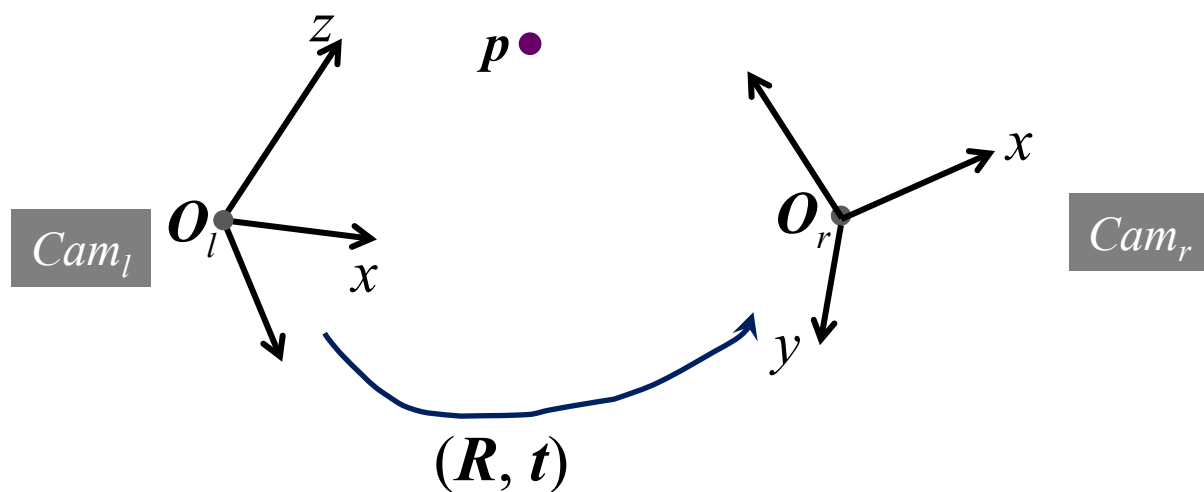
Calibrate the extrinsics between two cameras, i.e., the **Euclidean transformation** between the two camera coordinate systems

Our current focus

When calibrating the extrinsics, we suppose that the intrinsics of the two cameras are known



Calibration of a physical binocular system



A physical binocular system

- ✓ Cam_l and Cam_r are the two cameras
- ✓ The camera coordinate system of Cam_l is C_l
- ✓ The camera coordinate system of Cam_r is C_r

The extrinsics of such a physical binocular system are

$$R \in \mathbb{R}^{3 \times 3} \left(RR^T = I, \det(R) = 1 \right), t \in \mathbb{R}^{3 \times 1}$$

For a 3D point p , if its coordinate w.r.t. to C_l is p_l and its coordinate w.r.t. C_r is p_r , then p_l and p_r should satisfy,

$$p_r = Rp_l + t$$

By binocular calibration, our purpose is to get R and t



Calibration of a physical binocular system

For binocular system calibration, we also need to resort to the checkerboard



Collect m pairs of binocular images using the binocular suite

$$\longrightarrow \{I_{li}, I_{ri}\}_{i=1}^m$$



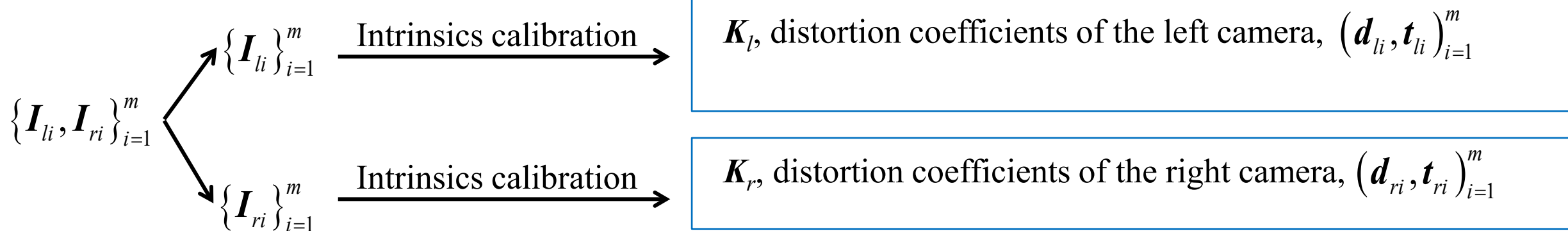
the left image

the right image

An example of a binocular pair



Calibration of a physical binocular system



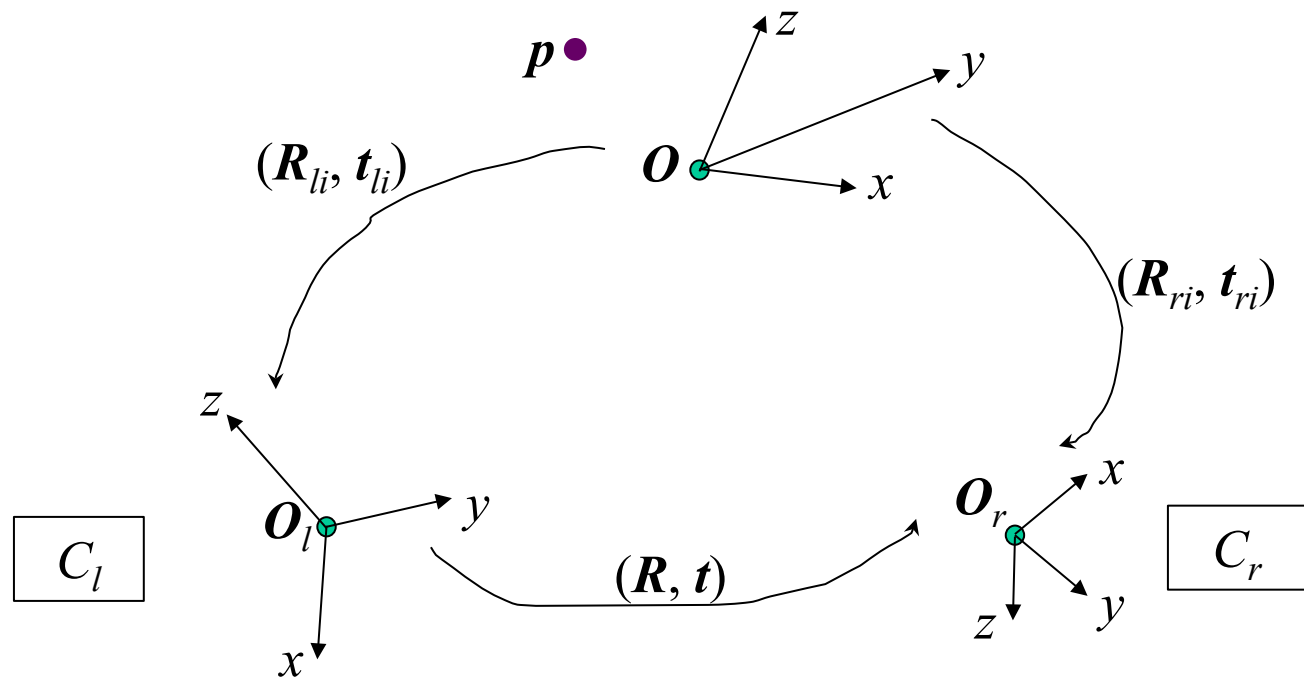
(d_{li}, t_{li}) is the left camera pose w.r.t. the world coordinate system induced by the calibration board when taking the i -th binocular pair; (d_{ri}, t_{ri}) is the right camera pose w.r.t. the world coordinate system induced by the calibration board when taking the i -th binocular pair

Based on $(R_{li} \triangleq \mathcal{R}(d_{li}), t_{li})$ and $(R_{ri} \triangleq \mathcal{R}(d_{ri}), t_{ri})$ ($\mathcal{R}(\cdot)$ is the Rodrigues operator taking an axis-angle to the corresponding rotation matrix), the extrinsics of the binocular system R and t can be derived





Calibration of a physical binocular system



$$p_l = R_{li} p + t_{li} \quad (3)$$



$$p = R_{li}^{-1} (p_l - t_{li}) \quad (6)$$

A byproduct we will use later

$$R_{ri} = R R_{li}, \quad t_{ri} = R t_{li} + t \quad (9)$$

$$p_r = R_{ri} p + t_{ri} \quad (4)$$

$$p_r = R p_l + t \quad (5)$$

$$\begin{aligned} p_r &= R_{ri} (R_{li}^{-1} (p_l - t_{li})) + t_{ri} \\ &= R_{ri} R_{li}^{-1} p_l + (t_{ri} - R_{ri} R_{li}^{-1} t_{li}) \end{aligned} \quad (7)$$

By comparing (5) and (7), we have

$$R = R_{ri} R_{li}^{-1}, \quad t = t_{ri} - R_{ri} R_{li}^{-1} t_{li} \quad (8)$$



Only the extrinsics of two cameras w.r.t the WCS when taking one binocular pair is used; not robust



We need to use all the m sets of binocular pairs of the calibration board $\{I_{li}, I_{ri}\}_{i=1}^m$ to estimate R and t





Calibration of a physical binocular system

Using the sum of the reprojection errors of the checker-board points on the left and right images as the cost:

$$\theta^* = \arg \min_{\theta} \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^n \left\{ \left\| \mathbf{K}_l \mathcal{D}_l \left\{ \frac{1}{z_{clij}} [\mathbf{R}_{li} \ \mathbf{t}_{li}]_{3 \times 4} \mathbf{p}_j \right\} - \mathbf{u}_{lij} \right\|_2^2 + \left\| \mathbf{K}_r \mathcal{D}_r \left\{ \frac{1}{z_{crij}} [\mathbf{R}_{ri} \ \mathbf{t}_{ri}]_{3 \times 4} \mathbf{p}_j \right\} - \mathbf{u}_{rij} \right\|_2^2 \right\}$$



According to Eq. (9)

$$\theta^* = \arg \min_{\theta} \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^n \left\{ \left\| \mathbf{K}_l \mathcal{D}_l \left\{ \frac{1}{z_{clij}} [\mathcal{R}(\mathbf{d}_{li}) \ \mathbf{t}_{li}]_{3 \times 4} \mathbf{p}_j \right\} - \mathbf{u}_{lij} \right\|_2^2 + \left\| \mathbf{K}_r \mathcal{D}_r \left\{ \frac{1}{z_{crij}} [\mathcal{R}(\mathbf{d}) \mathcal{R}(\mathbf{d}_{li}) \ \mathcal{R}(\mathbf{d}) \mathbf{t}_{li} + \mathbf{t}]_{3 \times 4} \mathbf{p}_j \right\} - \mathbf{u}_{rij} \right\|_2^2 \right\} \quad (10)$$

where $\theta = (\mathbf{d}, \mathbf{t}, \{\mathbf{d}_{li}\}_{i=1}^m, \{\mathbf{t}_{li}\}_{i=1}^m)$ are the parameters that need to be optimized

$\mathbf{u}_{lij}$ is the projection of $\mathbf{p}_j$ (a cross-point on the calibration board) onto the left image of the i th binocular pair

$\mathbf{u}_{rij}$ is the projection of $\mathbf{p}_j$ onto the right image of the i th binocular pair

$(\mathcal{R}(\mathbf{d}_{li}), \mathbf{t}_{li})$ are the extrinsics of the left camera w.r.t. to the world coordinate system when taking the i th binocular pair



Calibration of a physical binocular system

$$\mathbf{f}_{ij}(\boldsymbol{\theta}) \triangleq \begin{pmatrix} \mathbf{K}_l \mathcal{D}_l \left\{ \frac{1}{z_{clij}} [\mathcal{R}(\mathbf{d}_{li}) \mathbf{t}_{li}]_{3 \times 4} \mathbf{p}_j \right\} - \mathbf{u}_{lij} \\ \mathbf{K}_r \mathcal{D}_r \left\{ \frac{1}{z_{crij}} [\mathcal{R}(\mathbf{d}) \mathcal{R}(\mathbf{d}_{li}) \mathcal{R}(\mathbf{d}) \mathbf{t}_{li} + \mathbf{t}]_{3 \times 4} \mathbf{p}_j \right\} - \mathbf{u}_{rij} \end{pmatrix}_{4 \times 1}, i = 1, 2, \dots, m, j = 1, 2, \dots, n \quad \mathbf{f}(\boldsymbol{\theta}) \triangleq \left[\left(\mathbf{f}_{11}^T(\boldsymbol{\theta}), \mathbf{f}_{12}^T(\boldsymbol{\theta}), \dots, \mathbf{f}_{mn}^T(\boldsymbol{\theta}) \right)^T \right]_{4mn \times 1}$$



The problem of Eq. (10) can be formulated as $\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \frac{1}{2} \mathbf{f}^T(\boldsymbol{\theta}) \mathbf{f}(\boldsymbol{\theta})$

which is a typical nonlinear least-squares problem and can be used by using techniques introduced in Chapter 9

To derive the Jacobian matrix $\mathbf{J}(\boldsymbol{\theta})$ of $\mathbf{f}(\boldsymbol{\theta})$, refer to Chapter 10

For the initialization of $\boldsymbol{\theta}$, refer to the next page



Calibration of a physical binocular system

$$\theta = (d, t, \{d_{li}\}_{i=1}^m, \{t_{li}\}_{i=1}^m)$$

The initial values for $\{d_{li}, t_{li}\}_{i=1}^m$ can be obtained from the intrinsics calibration process for the left camera using $\{I_{li}\}_{i=1}^m$

According to Eq. 8, from the k th binocular pair, we can have a valid estimation for the initial value of (d, t) as (d_k, t_k)



The median value of $\{d_k\}_{k=1}^m \Rightarrow d$'s initial value

The median value of $\{t_k\}_{k=1}^m \Rightarrow t$'s initial value



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Establishment of the rectified binocular system

When the extrinsics of a physical binocular system are known, its associated rectified binocular system could be derived

- 1) Establish axis-aligned camera coordinate systems $C_{align-l}$ and $C_{align-r}$ from the original physical camera coordinate systems
- 2) By rotating $C_{align-l}$ and $C_{align-r}$, get the rectified camera coordinate system C_{rect-l} and C_{rect-r}
- 3) Determine the intrinsics of the rectified binocular system



Establishment of the rectified binocular system

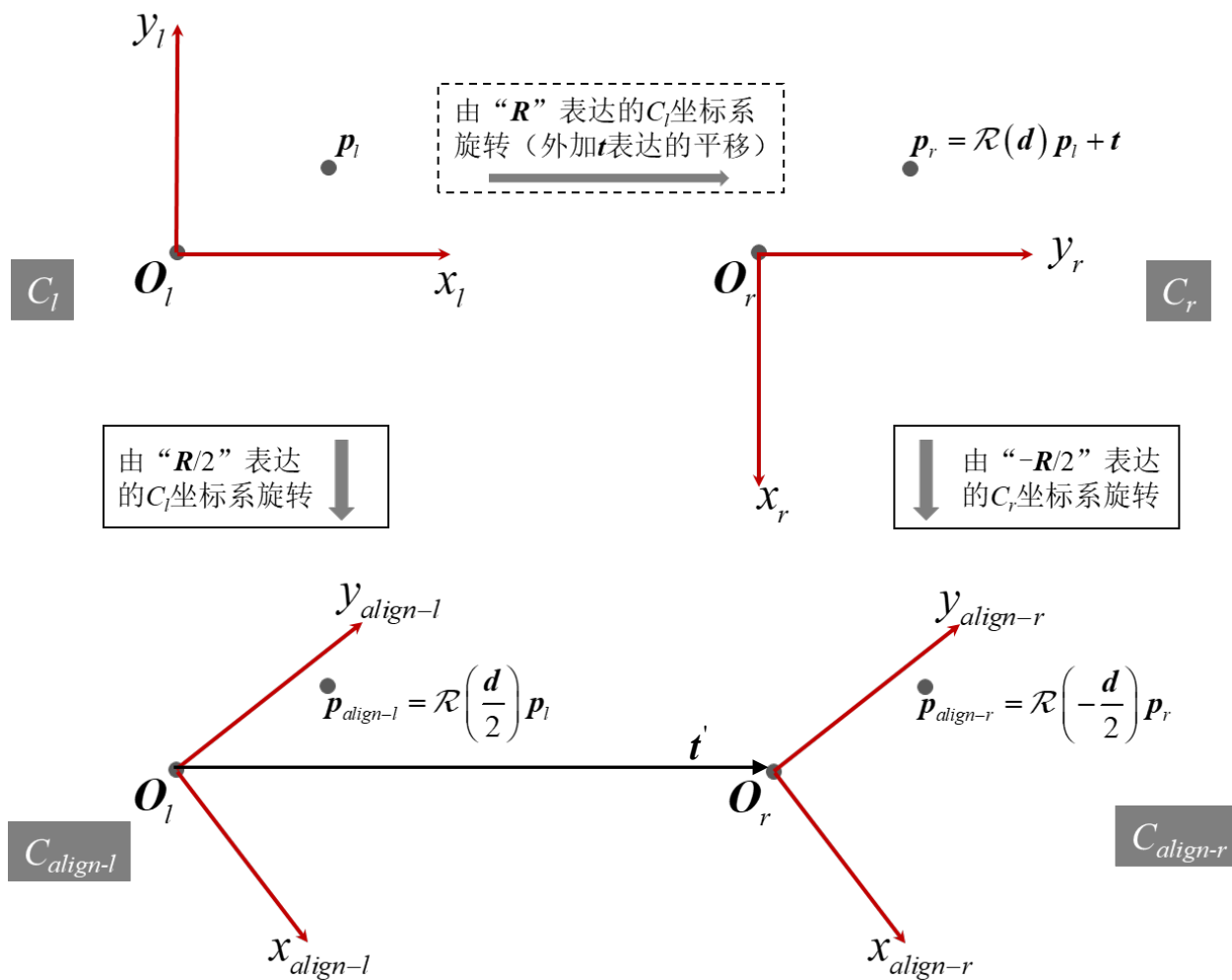
Notations

	Coordinate system of the left camera (coordinate of a spatial point w.r.t. this camera)	Coordinate system of the right camera (coordinate of a spatial point w.r.t. this camera)
physical camera systems	$C_l(\mathbf{p}_l)$	$C_r(\mathbf{p}_r)$
axis-aligned camera systems	$C_{align-l}(\mathbf{p}_{align-l})$	$C_{align-r}(\mathbf{p}_{align-r})$
rectified camera systems	$C_{rect-l}(\mathbf{p}_{rect-l})$	$C_{rect-r}(\mathbf{p}_{rect-r})$



Establishment of the rectified binocular system

Step 1: Establish axis-aligned camera coordinate systems $C_{align-l}$ and $C_{align-r}$



$$p_{align-l} = \mathcal{R}\left(\frac{d}{2}\right)p_l \quad (11), \quad p_{align-r} = \mathcal{R}\left(-\frac{d}{2}\right)p_r \quad (12)$$

$$p_r = Rp_l + t \quad (\text{multiply } \mathcal{R}\left(-\frac{d}{2}\right))$$

$$\mathcal{R}\left(-\frac{d}{2}\right)p_r = \mathcal{R}\left(-\frac{d}{2}\right)\mathcal{R}(d)p_l + \mathcal{R}\left(-\frac{d}{2}\right)t = \mathcal{R}\left(\frac{d}{2}\right)p_l + \mathcal{R}\left(-\frac{d}{2}\right)t \quad (13)$$

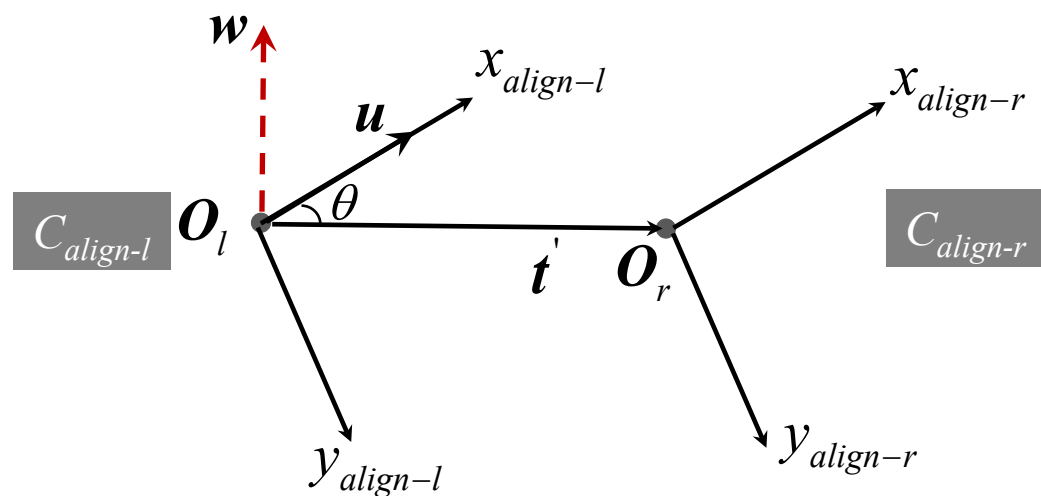
$$p_{align-r} = p_{align-l} + \underbrace{\mathcal{R}\left(-\frac{d}{2}\right)t}_{\overrightarrow{O_r O_l}}$$

$$\text{w.r.t. } C_{align-l}, \quad t' \triangleq \overrightarrow{O_l O_r} = -\mathcal{R}\left(-\frac{d}{2}\right)t$$



Establishment of the rectified binocular system

Step 2: Get the rectified camera coordinate system C_{rect-l} and C_{rect-r} by rotating $C_{align-l}$ and $C_{align-r}$ to make their x -axes coincide with the baseline vector $\overline{o_l o_r}$



w.r.t. $C_{align-l}$,

$u = (1, 0, 0)^T$ (actually the x -axis of $C_{align-l}$)

$$w = \frac{t' \times u}{\|t' \times u\|_2}, \theta = \arccos \left(\frac{t' \cdot u}{\|t'\|_2 \|u\|_2} \right)$$

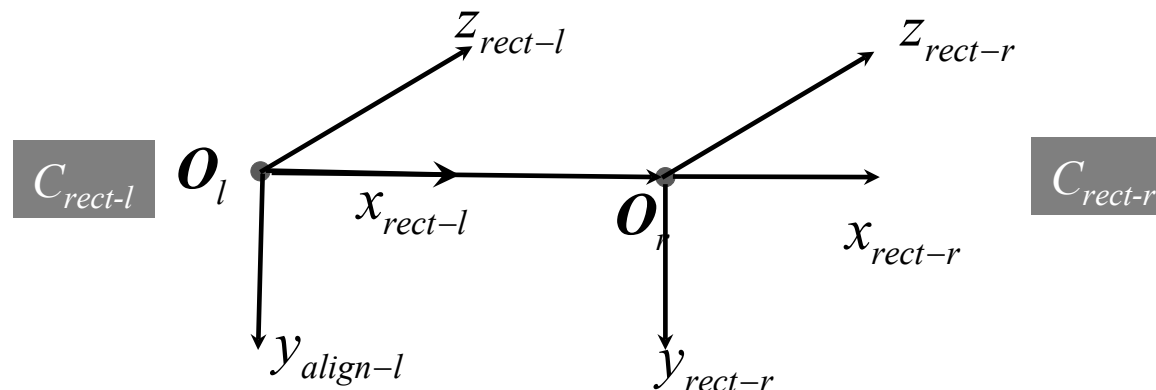
By rotating $C_{align-l}$ around w the angle $-\theta$, we can get C_{rect-l} ; Since $C_{align-r}$ and $C_{align-l}$ are axis-aligned, by applying the same rotation to $C_{align-r}$, we get C_{rect-r}





Establishment of the rectified binocular system

Step 2: Get the rectified camera coordinate system C_{rect-l} and C_{rect-r} by rotating $C_{align-l}$ and $C_{align-r}$ to make their x -axes coincide with the baseline vector $\overline{o_l o_r}$



For a point, if its coordinate w.r.t. $C_{align-l}$ is $\mathbf{p}_{align-l}$ and its coordinate w.r.t. C_{rect-l} is $\mathbf{p}_{rect-l}$, we have,

$$\mathbf{p}_{rect-l} = \mathcal{R}(\mathbf{w}\theta) \mathbf{p}_{align-l} \quad (14)$$

For a point, if its coordinate w.r.t. $C_{align-r}$ is $\mathbf{p}_{align-r}$ and its coordinate w.r.t. C_{rect-r} is $\mathbf{p}_{rect-r}$, we have,

$$\mathbf{p}_{rect-r} = \mathcal{R}(\mathbf{w}\theta) \mathbf{p}_{align-r} \quad (15)$$



Establishment of the rectified binocular system

Step 2: Get the rectified camera coordinate system C_{rect-l} and C_{rect-r} by rotating $C_{align-l}$ and $C_{align-r}$ to make their x -axes coincide with the baseline vector $\overline{o_l o_r}$

Consider a spatial point. Suppose its coordinate w.r.t. C_l is $\mathbf{p}_l$ and its coordinate w.r.t. C_{rect-l} is $\mathbf{p}_{rect-l}$. What is the relationship between $\mathbf{p}_l$ and $\mathbf{p}_{rect-l}$?

$$\left. \begin{array}{l} \text{According to Eq. (11)} \quad \mathbf{p}_{align-l} = \mathcal{R}\left(\frac{d}{2}\right) \mathbf{p}_l \\ \text{According to Eq. (14)} \quad \mathbf{p}_{rect-l} = \mathcal{R}(\mathbf{w}\theta) \mathbf{p}_{align-l} \end{array} \right\} \mathbf{p}_{rect-l} = \mathcal{R}(\mathbf{w}\theta) \mathbf{p}_{align-l} = \mathcal{R}(\mathbf{w}\theta) \mathcal{R}\left(\frac{d}{2}\right) \mathbf{p}_l$$

$\mathbf{R}_{rect-l}$

Consider a spatial point. Suppose its coordinate w.r.t. C_r is $\mathbf{p}_r$ and its coordinate w.r.t. C_{rect-r} is $\mathbf{p}_{rect-r}$. What is the relationship between $\mathbf{p}_r$ and $\mathbf{p}_{rect-r}$?

$$\left. \begin{array}{l} \text{According to Eq. (12)} \quad \mathbf{p}_{align-r} = \mathcal{R}\left(-\frac{d}{2}\right) \mathbf{p}_r \\ \text{According to Eq. (15)} \quad \mathbf{p}_{rect-r} = \mathcal{R}(\mathbf{w}\theta) \mathbf{p}_{align-r} \end{array} \right\} \mathbf{p}_{rect-r} = \mathcal{R}(\mathbf{w}\theta) \mathbf{p}_{align-r} = \mathcal{R}(\mathbf{w}\theta) \mathcal{R}\left(-\frac{d}{2}\right) \mathbf{p}_r$$

$\mathbf{R}_{rect-r}$



Establishment of the rectified binocular system

Step 2: Get the rectified camera coordinate system C_{rect-l} and C_{rect-r} by rotating $C_{align-l}$ and $C_{align-r}$ to make their x -axes coincide with the baseline vector $\overrightarrow{o_l o_r}$

$$\mathbf{p}_{rect-l} = \mathcal{R}(w\theta) \mathbf{p}_{align-l} = \mathcal{R}(w\theta) \mathcal{R}\left(\frac{d}{2}\right) \mathbf{p}_l \quad (16)$$

$$\mathbf{p}_{rect-r} = \mathcal{R}(w\theta) \mathbf{p}_{align-r} = \mathcal{R}(w\theta) \mathcal{R}\left(-\frac{d}{2}\right) \mathbf{p}_r \quad (17)$$

 $\mathbf{R}_{rect-l}$ $\mathbf{R}_{rect-r}$

Rectification rotation matrix

$\mathbf{R}_{rect-l}$ can transform a point's coordinate under C_l to C_{rect-l} while $\mathbf{R}_{rect-r}$ can transform a point's coordinate under C_r to C_{rect-r}



Establishment of the rectified binocular system

Step 2: Get the rectified camera coordinate system C_{rect-l} and C_{rect-r} by rotating $C_{align-l}$ and $C_{align-r}$ to make their x -axes coincide with the baseline vector

Let's consider the extrinsics of the rectified binocular system, i.e., the relationship between C_{rect-l} and C_{rect-r}

According to Eq. (13),

$$\mathcal{R}\left(-\frac{d}{2}\right)p_r = \mathcal{R}\left(\frac{d}{2}\right)p_l + \mathcal{R}\left(-\frac{d}{2}\right)t \quad (13)$$



Multiply $\mathcal{R}(w\theta)$

$$\mathcal{R}(w\theta)\mathcal{R}\left(-\frac{d}{2}\right)p_r = \mathcal{R}(w\theta)\mathcal{R}\left(\frac{d}{2}\right)p_l + \mathcal{R}(w\theta)\mathcal{R}\left(-\frac{d}{2}\right)t$$



According to Eq. (16) and Eq. (17)

$$p_{rect-r} = p_{rect-l} + R_{rect-r}t$$

So, the extrinsics of the rectified binocular system is $(I, t_{rect} \triangleq R_{rect-r}t)$



Theoretically speaking, only the first element of t_{rect} is nonzero

The length of the baseline $b = \|\overrightarrow{O_l O_r}\| = \|t_{rect}\|_2 = \|R_{rect-r}t\|_2 = \|t\|_2$



Establishment of the rectified binocular system

Step 3: Determine the intrinsics of the rectified binocular system

We know that for a rectified binocular system, the two distortion coefficient vectors are zeros and the two cameras share the same intrinsic matrix. Thus, we only need to specify $\mathbf{K}_{rect}$,

$$\mathbf{K}_{rect} = \begin{bmatrix} f_{rect} & 0 & c_{rect-x} \\ 0 & f_{rect} & c_{rect-y} \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Suppose the intrinsics of the left and right physical cameras are $\mathbf{K}_l = \begin{bmatrix} f_{x-l} & 0 & c_{x-l} \\ 0 & f_{y-l} & c_{y-l} \\ 0 & 0 & 1 \end{bmatrix}$ and $\mathbf{K}_r = \begin{bmatrix} f_{x-r} & 0 & c_{x-r} \\ 0 & f_{y-r} & c_{y-r} \\ 0 & 0 & 1 \end{bmatrix}$

The elements in $\mathbf{K}_{rect}$ can be specified as,

$$f_{rect} \triangleq \min(f_{x-l}, f_{y-l}, f_{x-r}, f_{y-r})$$

$$c_{rect-x} \triangleq (c_{x-l} + c_{x-r}) / 2$$

$$c_{rect-y} \triangleq (c_{y-l} + c_{y-r}) / 2$$



Establishment of the rectified binocular system

Please verify that, after steps 1~3, we really get a rectified binocular system

At runtime, when capturing a physical binocular pair, how to get its associated rectified binocular pair? i.e, how to project a physical binocular pair to a rectified binocular pair?



Outline

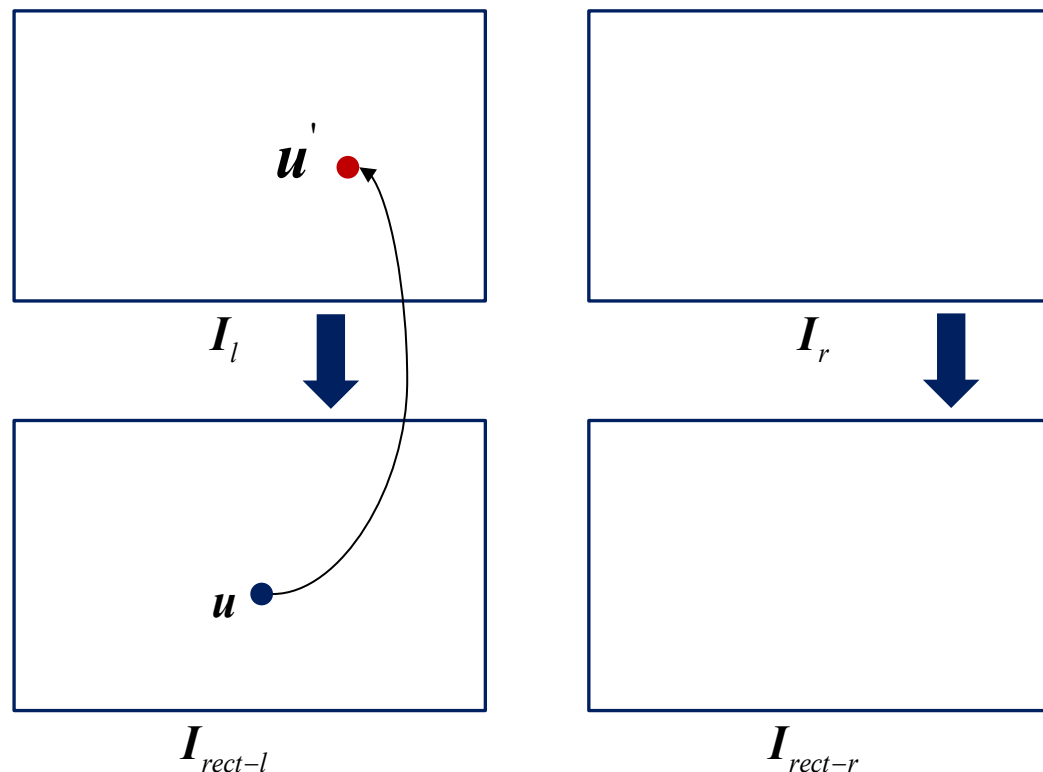
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Obtaining the rectified binocular pair from a physical binocular pair

physical binocular pair (given)
captured by two physical
cameras Cam_l and Cam_r

rectified binocular pair (need to
be determined) of two rectified
cameras Cam_{rect-l} and Cam_{rect-r}



The core issue is: for a point u (normalized homogeneous form) on I_{rect-l} , how to determine its corresponding point u' on I_l ?

Then, using $I_{rect-l}(u) \leftarrow I_l(u')$, I_{rect-l} can be determined



Obtaining the rectified binocular pair from a physical binocular pair

$\mathbf{u}'$ can be determined by,

$$\mathbf{u}' = \mathbf{K}_l \cdot \mathcal{D}_l \left\{ \frac{\mathbf{R}_{rect-l}^{-1} \mathbf{K}_{rect}^{-1} \mathbf{u}}{z_{cam_l}} \right\}$$

$\mathbf{K}_{rect}^{-1} \mathbf{u}$: $\mathbf{u}$'s corresponding point on the normalized retinal plane of the left rectified camera Cam_{rect-l} ; it can also be regarded as 3D point $\mathbf{p}$ w.r.t. C_{rect-l}

$\mathbf{R}_{rect-l}^{-1} \mathbf{K}_{rect}^{-1} \mathbf{u}$: $\mathbf{p}$'s coordinate w.r.t. C_l

$\frac{\mathbf{R}_{rect-l}^{-1} \mathbf{K}_{rect}^{-1} \mathbf{u}}{z_{cam_l}}$: $\mathbf{p}$'s projection on the normalized retinal plane of the left physical camera Cam_l

$\mathcal{D}_l \left\{ \frac{\mathbf{R}_{rect-l}^{-1} \mathbf{K}_{rect}^{-1} \mathbf{u}}{z_{cam_l}} \right\}$: $\mathbf{p}$'s distorted projection on the normalized retinal plane of Cam_l where $\mathcal{D}_l(.)$ is the distortion operator of Cam_l

$\mathbf{K}_l \cdot \mathcal{D}_l \left\{ \frac{\mathbf{R}_{rect-l}^{-1} \mathbf{K}_{rect}^{-1} \mathbf{u}}{z_{cam_l}} \right\}$: $\mathbf{p}$'s image (pixel position) on $\mathbf{I}_l$

Using a similar procedure, $\mathbf{I}_{rect-r}$ can be derived based on $\mathbf{I}_r$



Outline

- The flowchart to build a binocular system
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- Practice



Stereo matching and disparity map computation for a rectified binocular pair

When the extrinsics and the intrinsics of a physical binocular system are known, its associated rectified binocular system can be constructed

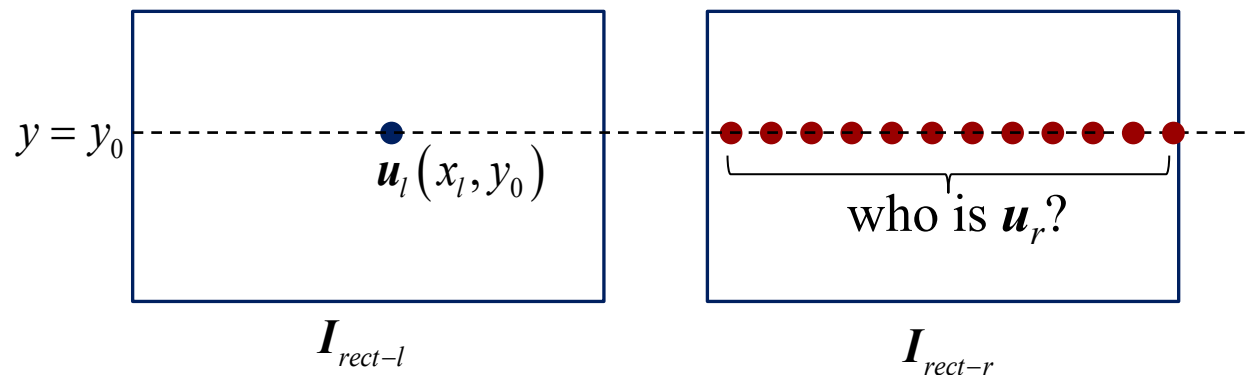
To get the corresponding depth map to the left rectified image I_{rect-l} , we need to at first extract its disparity map



We need to know the matching relationships among pixels on I_{rect-l} and I_{rect-r}

For a point $\mathbf{u}_l(x_l, y_0)$ on I_{rect-l} , how to find the matching pixel $\mathbf{u}_r$ on I_{rect-r} ?

Since I_{rect-l} and I_{rect-r} are row-aligned, $\mathbf{u}_r$ can only happen on the y_0 row of I_{rect-r}





Stereo matching and disparity map computation for a rectified binocular pair

On I_{rect-l} , take a patch A round $\mathbf{u}_l$. Traverse each position of y_0 row on I_{rect-r} . For the position (x_i, y_0) on I_{rect-r} , take a patch B_i around (x_i, y_0)

Figuring out the position i^* whose local patch has the least distance to A ,

$$i^* = \arg \min_{i=1,2,\dots,n} d(A, B_i)$$

where $d(.,.)$ is a function returning the distance between two vectors

(x_{i^*}, y_0) is $\mathbf{u}_r$ we want to find, i.e., $\mathbf{u}_r = (x_{i^*}, y_0)$

Note: in implementation, usually the matching is not conducted on the raw images but on the partial derivative images, for example, the results of Sobel operators; there are also some algorithms to postprocess the raw disparity map to make it more robust, such as the weighted least-squares method^[1]

[1] MIN D, CHOI S, LU J, et al. Fast global image smoothing based on weighted least squares[J]. IEEE Trans. Image Processing, 2014, 23(12): 5638-5653.



Stereo matching and disparity map computation for a rectified binocular pair



The raw disparity map



The disparity map with weighted least-squares postprocessing



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3D reconstruction based on the disparity map in the rectified binocular system

With the rectified binocular system, suppose I_{rect-l} is its left image and $\mathbf{u}(x, y)$ is one pixel on I_{rect-l} and its disparity is known as d . The 3D point in space corresponding to $\mathbf{u}$ is $\mathbf{p}$. Now let's compute the coordinate of $\mathbf{p}$ w.r.t. the coordinate system C_{rect-l}

Suppose $\mathbf{x}_n = (x_n, y_n, 1)^T$ is the point on the normalized retinal plane of Cam_{rect-l} corresponding to $\mathbf{u}$. We can have

$$\begin{bmatrix} f_{rect} & 0 & c_{rect-x} \\ 0 & f_{rect} & c_{rect-y} \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x_n \\ y_n \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \Rightarrow \mathbf{x}_n = \begin{pmatrix} x_n \\ y_n \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{x - c_{rect-x}}{f_{rect}} \\ \frac{y - c_{rect-y}}{f_{rect}} \\ 1 \end{pmatrix}$$

According to Eq. (2), $\mathbf{p}$'s depth value (z value) w.r.t. C_{rect-l} is $z = \frac{f_{rect}}{d} b$





3D reconstruction based on the disparity map in the rectified binocular system

$\mathbf{p}$'s coordinate w.r.t. C_{rect-l} is,

$$\mathbf{z}\mathbf{x}_n = \frac{f_{rect}b}{d} \cdot \begin{pmatrix} \frac{x - c_{rect-x}}{f_{rect}} \\ \frac{y - c_{rect-y}}{f_{rect}} \\ 1 \end{pmatrix} = \begin{pmatrix} (x - c_{rect-x})\frac{b}{d} \\ (y - c_{rect-y})\frac{b}{d} \\ f_{rect}\frac{b}{d} \end{pmatrix}$$

The homogeneous form of $\mathbf{p}$'s position w.r.t. C_{rect-l} is,

$$\left((x - c_{rect-x})\frac{b}{d}, (y - c_{rect-y})\frac{b}{d}, f_{rect}\frac{b}{d}, 1 \right)^T \xrightarrow{\text{or } \times \frac{d}{b}} \left((x - c_{rect-x}), (y - c_{rect-y}), f_{rect}, \frac{d}{b} \right)^T$$

Notice that,

$$\begin{pmatrix} x - c_{rect-x} \\ y - c_{rect-y} \\ f_{rect} \\ \frac{d}{b} \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & -c_{rect-x} \\ 0 & 1 & 0 & -c_{rect-y} \\ 0 & 0 & 0 & f_{rect} \\ 0 & 0 & \frac{1}{b} & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ d \\ 1 \end{pmatrix}$$

Denote $\mathbf{Q} \triangleq \begin{bmatrix} 1 & 0 & 0 & -c_{rect-x} \\ 0 & 1 & 0 & -c_{rect-y} \\ 0 & 0 & 0 & f_{rect} \\ 0 & 0 & \frac{1}{b} & 0 \end{bmatrix}$





3D reconstruction based on the disparity map in the rectified binocular system

$$\mathbf{Q} \begin{pmatrix} x \\ y \\ d \\ 1 \end{pmatrix} \triangleq \begin{pmatrix} x_h \\ y_h \\ z_h \\ w_h \end{pmatrix}$$

is the 3D point (in homogeneous form) w.r.t. C_{rect-l} corresponding to the pixel $\mathbf{u}$ of the left rectified image $\mathbf{I}_{rect-l}$

Its normal 3D coordinate is,

$$\begin{pmatrix} \frac{x_h}{w_h} \\ \frac{y_h}{w_h} \\ \frac{z_h}{w_h} \end{pmatrix}$$

For any pixel $\mathbf{u}(x, y)$ on the left image $\mathbf{I}_{rect-l}$ of the rectified binocular system, if its disparity is d , then, the spatial point $\mathbf{p}$ (corresponding to $\mathbf{u}$)'s coordinate w.r.t. the camera coordinate system of the left rectified camera is,

$$\mathbf{Q}(x, y, d, 1)^T$$



3D reconstruction based on the disparity map in the rectified binocular system

An example

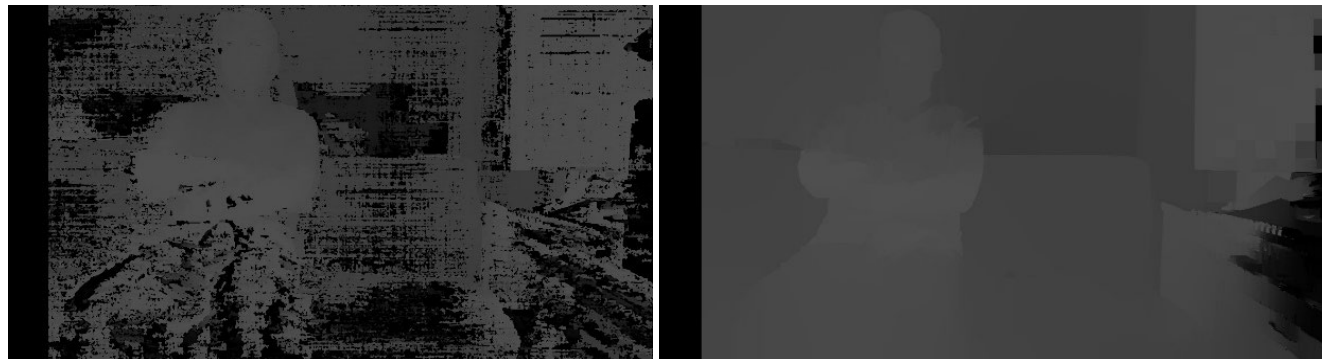
Physical binocular pair



Rectified binocular pair
(they are row-aligned)



raw depth map



depth map with
weighted least-squares
postprocessing



3D reconstruction based on the disparity map in the rectified binocular system

An example



The RGB-D point cloud



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Theory to code

The screenshot shows the GitHub interface for the repository 'csLinZhang / CVBook'. The breadcrumb path is 'CVBook / chapter-17-stereo'. The left sidebar shows a list of files and folders, including 'chapter-04-feature detection and...', 'chapter-06-homography estimati...', 'chapter-09-Levenberg-Marquardt', 'chapter-10-imaging model and in...', 'chapter-11-bird-eye view', 'chapter-14-SVM', and 'chapter-15-YOLO'. The main content area shows the 'chapter-17-stereo' directory with a table of files and folders. The 'matlab-calibration-theory' folder is highlighted with a red box.

Name	Last commit message
..	
matlab-calibration-theory	Add files via upload
matlab-stereo	Add files via upload
stereoCPP	Add files via upload

Matlab source code to perform binocular system calibration and stereo rectification



Construct a binocular stereo system—Matlab



Fix two cameras on a plane to get a physical binocular system



Construct a binocular stereo system—Matlab

github.com/csLinZhang/CVBook/tree/main/chapter-17-stereo/matlab-stereo

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Code Issues 4 Pull requests Actions Projects Security Insights

Files

main

Go to file

- load_stereo_calib_files.m
- project_points2.m
- rect_index.m
- rectify_stereo_pair.m
- rigid_motion.m
- rodrigues.m
- saving_stereo_calib.m
- write_image.m
- matlab-stereo
 - imgsCollect

CVBook / chapter-17-stereo / matlab-stereo /

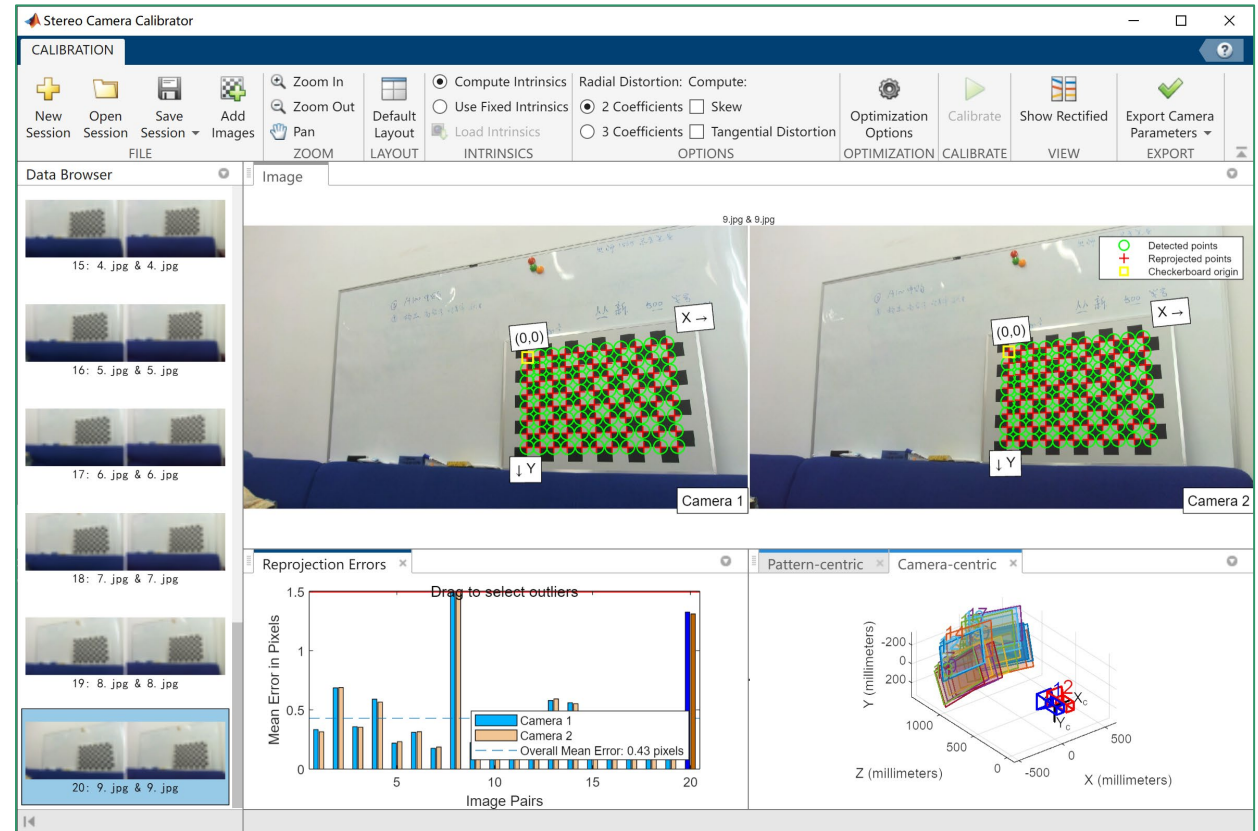
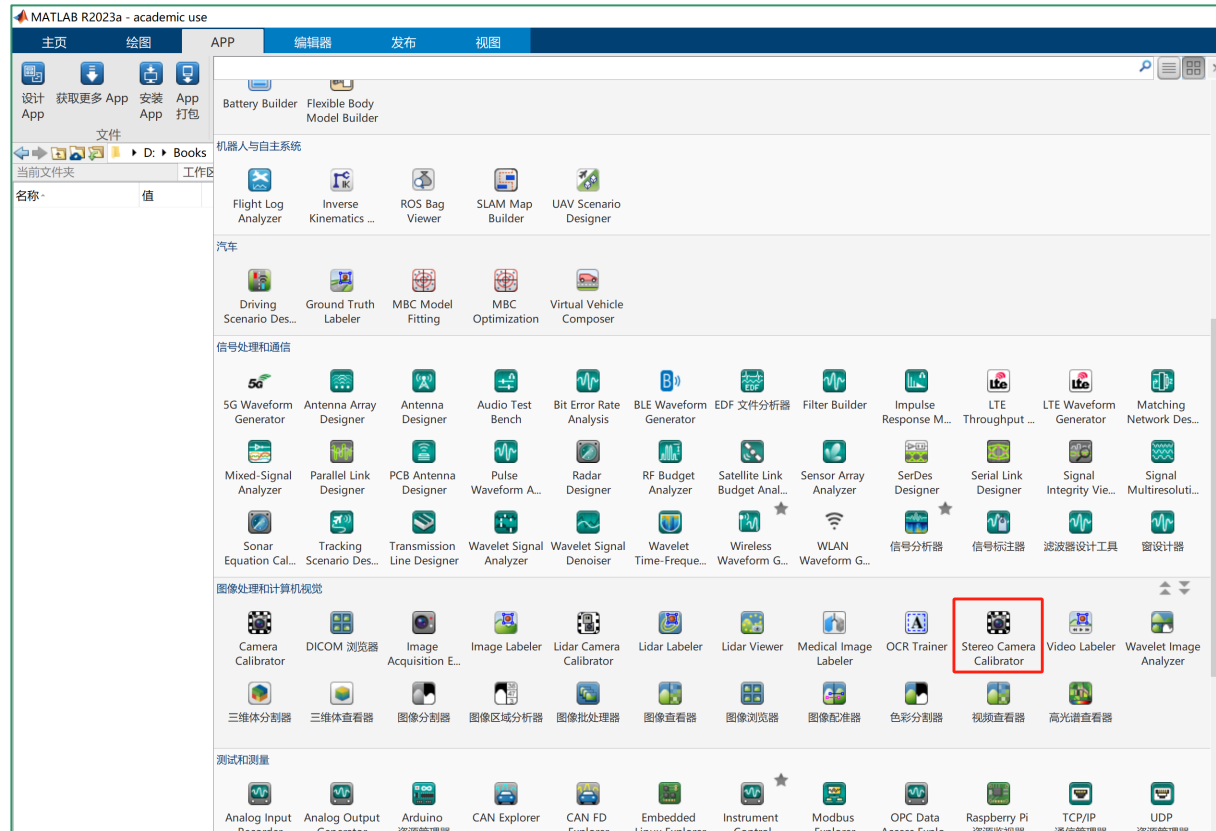
csLinZhang Add files via upload

Name	Last commit message
..	
imgsCollect	Add files via upload
stereo-imgs	Add files via upload
main.m	Add files via upload
result.ply	Add files via upload
stereoParams.mat	Add files via upload

A C++ program to collect binocular image pairs



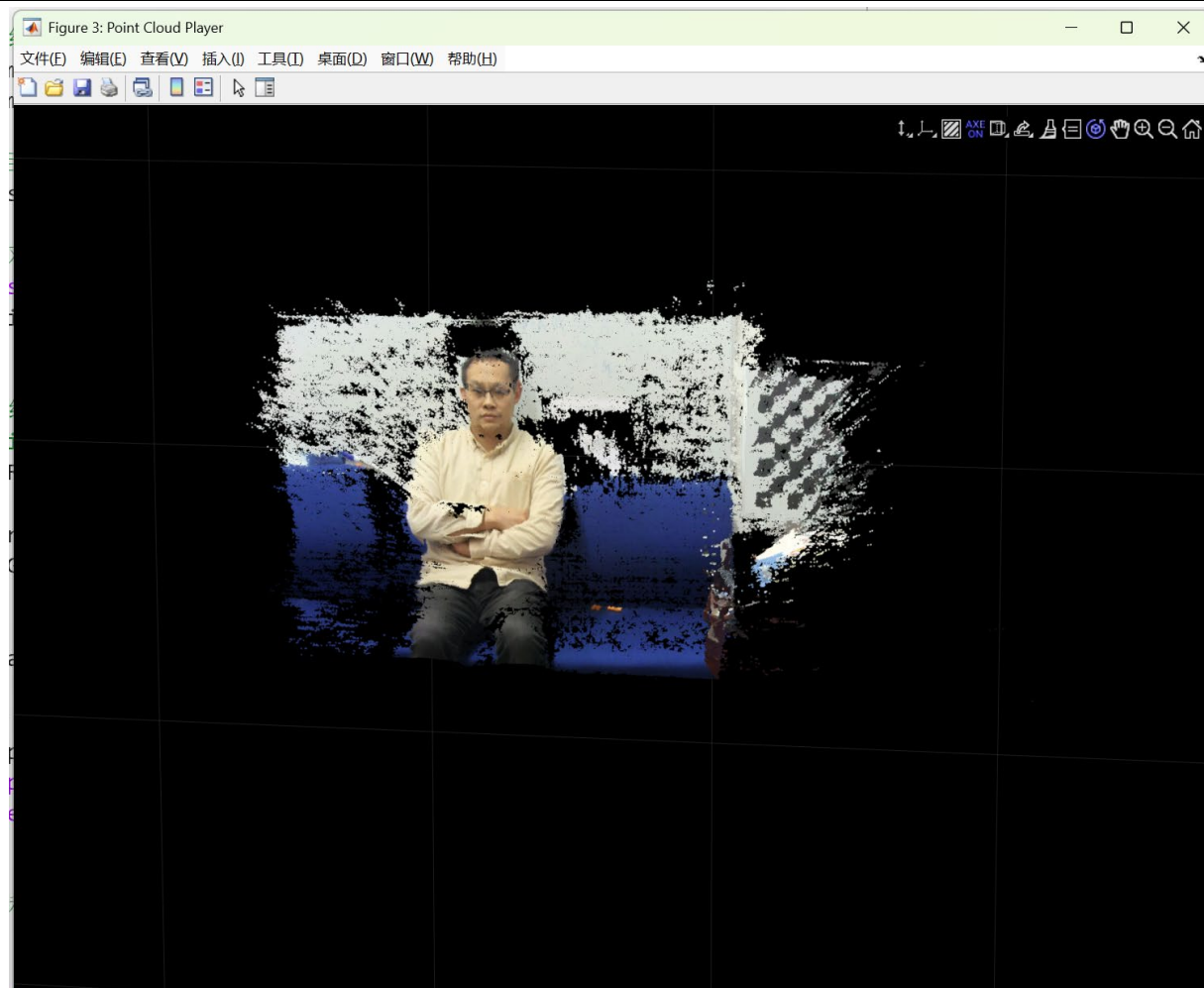
Construct a binocular stereo system—Matlab



Stereo Camera Calibrator can be used to calibrate a binocular system



Construct a binocular stereo system—Matlab



(Demo time)

When the binocular system has been calibrated, the rectified stereo pair can be obtained and then the point cloud can be reconstructed



Construct a binocular stereo system—OpenCV and C++

github.com/csLinZhang/CVBook/tree/main/chapter-17-stereo

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Go to file

- > chapter-04-feature detection and...
- > chapter-06-homography estimati...
- > chapter-09-Levenberg-Marquardt
- > chapter-10-imaging model and in...
- > chapter-11-bird-eye view
- > chapter-14-SVM
- > chapter-15-YOLO
- ✓ chapter-17-stereo
- ✓ matlab-calibration-theory
 - > stereo_example
- Calib_Results_stereo.mat

CVBook / chapter-17-stereo /

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Name	Last commit message
..	
matlab-calibration-theory	Add files via upload
matlab-stereo	Add files via upload
stereoCPP	Add files via upload



Construct a binocular stereo system—OpenCV and C++

- Environment
 - Visual Studio 2017 (or higher version), OpenCV-4.8.1+OpenCV_Contrib, Eigen 3.4, Pangolin-0.6
- Three “main” cpp files
 - **imgCollect.cpp**, to collect binocular pairs; it includes three main functions used to collect binocular pairs from checkerboards, to collect binocular pairs from ordinary scenes, and to collect binocular pair videos
 - **stereoCalib.cpp**, to perform binocular system calibration; it will output the intrinsic parameters of the two cameras to “intrinsic.yaml” and output the extrinsic parameters of the binocular system to “extrinsic.yaml”
 - **stereo3D.cpp**, to compute the rectified binocular pair, the disparity map, the depth map, and the 3D point cloud

(Demo time)

